

↳ Find out the history of Towers of Hanoi

↳ Today from the morning 7:30 AM to 4:30 PM list out various data structures in your surrounding

Q1) Towers of Hanoi

↳ Tower of Hanoi is a famous mathematical puzzle that is widely used in DSA to teach recursion

History

⇒ It was invented by French mathematician Edouard Lucas in 1883.

⇒ Lucas created the puzzle to explain mathematical and recursive concepts in an interesting way

⇒ It is also known as the Tower of Brahma because of a famous legend associated with it

The legend

According to the legend

⇒ In a temple in Hanoi, there are three rods and 64 golden disks

⇒ monks continuously move the disks from one rod to another following the puzzle's rules.

⇒ It is said that when they finish moving all 64 disks, the world will come to an end.

This is only a legend and not a historical fact.

Rules of Tower of Hanoi

↳ only one disk can be moved at a time.

↳ only the top disk of a rod can be moved.

↳ A larger disk cannot be placed on a smaller disk.

Importance in DSA

- ↳ It is one of the best ex of recursion
- ↳ It helps students understand Recursive thinking, Divide & Conquer Strategy
- Recursive function calls, Time Complexity analysis

Time Complexity

No of moves required $= 2^n - 1$

Time Complexity $= O(2^n)$

Q2)

<u>Activity</u>	<u>Datastructure</u>	<u>Activity</u>	<u>DOS</u>
Alarm notification	Queue	Lunch queue	Queue
Tooth brush holder	Array	College Campus map	Graph
Clothes in wardrobe	Stack	Family tree example in class	Tree
Hostel bath room queue	Queue	Browser tabs on laptop/phone	Stack
Breakfast plate	Array	Contacts on mobile	Hash table
Students walking to class	Linked list	Hostel Stair case	Stack
Classroom Seating	2D array (Matrix)	Hostel building floors	Tree
Teacher's attendance list	(Arraylist)	Road from hostel to college	Graph
College time table	Array	Gallery photos	Array
WhatsApp chat messages	Linked list (ordered sequence)	items	Array
Music play list	linked list		